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COURSE OVERVIEW — CHAPTER BY CHAPTER

1. **Introduction** — Modbus was written in 1979 and has not changed its core protocol since. This chapter establishes why that matters: the same register model running in 1980s PLCs ships unchanged in new devices today. Engineers who understand the history stop treating its quirks as bugs.
2. **Data Model** — Four register tables — coil, discrete input, input register, holding register — define every possible Modbus transaction. This chapter makes the address space automatic before function codes are introduced.
3. **RTU Mode** — Serial framing, CRC-16, RS-485 multi-drop wiring, and the silence intervals that separate frames. RTU is still the dominant implementation in field devices; engineers who skip this chapter struggle with every serial deployment.
4. **ASCII Mode** — The human-readable serial variant: twice the overhead of RTU for the same data. Covered because legacy systems still use it, and because understanding it clarifies exactly why RTU won.
5. **Modbus TCP** — MBAP header, transaction IDs, port 502, and how Modbus crossed to Ethernet without modifying its register model. This is why Modbus remains relevant on modern plant networks.
6. **Function Codes** — The 20 codes covering 95% of deployments: read coils, read/write registers, write multiple registers, and the diagnostics codes that only appear when a device is misbehaving.
7. **Exception Codes** — Illegal Function, Illegal Data Address, Illegal Data Value, Slave Device Failure. Engineers learn to read exception responses instead of assuming the master is at fault.
8. **Data Types** — 16-bit registers were not designed for floats, 32-bit integers, or strings. This chapter covers register-pair encoding, byte-order conventions, and the vendor-specific choices that turn a simple read into an hour of debugging.
9. **Troubleshooting** — Addressing offset errors, byte-order mismatches, poll-rate overloads, RS-485 termination faults, and the firewall that silently drops Modbus TCP. Field diagnosis using Wireshark and software masters.
10. **Security** — Modbus has no authentication, no encryption, and no access control by design. This chapter covers compensating controls: network segmentation, unidirectional gateways, and anomaly detection at the network layer.
11. **Practice Lab** — Five diagnostic exercises: mapping an unknown device's registers, tracing an Exception Code 02, configuring an RTU-to-TCP gateway, diagnosing a byte-order fault, and building a Wireshark filter to monitor a Modbus TCP master.

Certification Relevance

- **ISA CCST** — Modbus appears in industrial communications and data acquisition domains at all three CCST levels.
- **GICSP (GIAC)** — OT protocol knowledge and Modbus security posture appear throughout ICS security exam domains.
- **CompTIA CySA+** — Industrial network traffic analysis including Modbus TCP anomaly detection.

Next Steps

1. Complete Modbus before OPC UA — the data model concepts transfer directly.
2. Download ModRsim2 (free) and Modbus Poll (trial) to practice register reads against a live simulator.
3. Review IEC 61158 for the formal specification behind the protocol details.
4. Pursue ISA CCST Level I as the entry-level industrial automation credential.